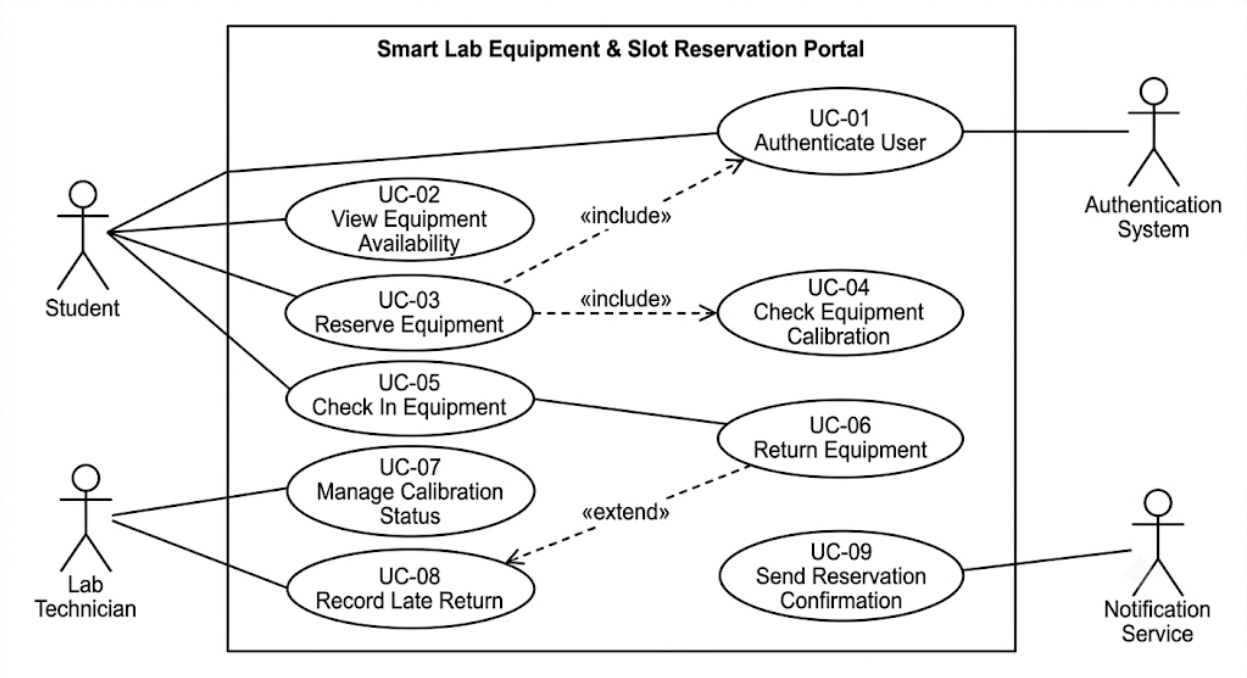
PRATHAM P MUDAKAVI
PES2UG24CS369
Section G
SE LAB 1

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow authenticated students to view real-time availability and reserve a maximum 2-hour slot for any calibrated equipment up to 7 days in advance.	High	**Pass:** Slot is locked and reservation token is generated. **Fail:** Double-booking occurs or reservation exceeds 2 hours.	Core reservation functionality and prevents equipment conflicts.
FR-002	Functional	The system shall allow lab technicians to update the calibration status of laboratory equipment.	High	**Pass:** Technician can mark equipment as calibrated or uncalibrated, and the updated status appears in the equipment listing.	Reservations must only be made for calibrated equipment.
FR-003	Functional	The system shall prevent students from reserving equipment that is already reserved for the selected time slot.	High	**Pass:** System rejects a conflicting reservation and displays the equipment's unavailable status.	Prevents double-booking of high-value equipment.

FR-004	Functional	The system shall allow students to check in and return reserved equipment against their active reservation.	Medium	**Pass:** Check-in and return are recorded with the reservation ID and timestamp.	Provides equipment usage tracking and supports late-return enforcement.
FR-005	Functional	The system shall record late equipment returns and apply the applicable disciplinary restriction to the student.	High	**Pass:** A return after the scheduled end time is recorded as late and the student's restriction is updated.	Enforces the disciplinary rules specified in the problem.
NFR-001	Nonfunctional	The system shall ensure that concurrent slot reservation requests are processed within 200 ms with strict database-level lock isolation to prevent race conditions.	High	**Pass:** Benchmarking under simulated concurrent load confirms processing within 200 ms and no double-booking occurs.	Ensures reliable reservations during high demand.
NFR-002	Nonfunctional	The system shall restrict equipment-management functions to authenticated lab technicians and reservation functions to authenticated students.	High	**Pass:** Unauthorized users attempting restricted functions are denied access.	Protects equipment data and prevents unauthorized reservations or status changes.



Use-Case Flow Specification – Reserve Equipment

Use Case ID: UC-03
Use Case Name: Reserve Equipment
Primary Actor: Student
Supporting Actors: Authentication System, Lab Technician, Notification Service

Preconditions

1. The student is authenticated in the Smart Lab Equipment & Slot Reservation Portal.
2. The required equipment is registered in the system.
3. The equipment has a valid calibration status.
4. The selected reservation time is within the next 7 days.
5. The requested reservation duration does not exceed 2 hours.

Postconditions

1. A valid reservation is created for the selected equipment and time slot.
2. The selected slot is locked to prevent double-booking.
3. A unique reservation token is generated.

4. The student receives a reservation confirmation.

Main Success Scenario

1. The student selects **View Equipment Availability**.
2. The system displays available laboratory equipment and their available time slots.
3. The student selects the required equipment and preferred time slot.
4. The system verifies that the equipment is calibrated.
5. The system verifies that the requested duration does not exceed 2 hours.
6. The system checks whether the selected time slot is available.
7. The system locks the selected slot using database-level locking.
8. The system creates the reservation.
9. The system generates a unique reservation token.
10. The system sends a reservation confirmation to the student.
11. The reservation process ends successfully.

Alternate Flow – Equipment Already Reserved

4a. Equipment is unavailable

1. The system detects that the selected equipment is already reserved for the requested time slot.
2. The system rejects the reservation request.
3. The system displays a message indicating that the selected slot is unavailable.
4. The student is prompted to select another available time slot.
5. The use case resumes from Step 3 of the Main Success Scenario.

Alternate Flow – Equipment Not Calibrated

4b. Equipment is not calibrated

1. The system determines that the selected equipment is not currently calibrated.
2. The system prevents the student from reserving the equipment.
3. The system displays a message indicating that the equipment is unavailable for reservation.
4. The student may select another calibrated equipment item.
5. The use case resumes from Step 2 of the Main Success Scenario.